

OSPF (Open Shortest Path First) – Lecture Notes

Introduction to OSPF

- OSPF stands for Open Shortest Path First.
- It is a link-state routing protocol used in IP networks.
- Uses Dijkstra's shortest path first (SPF) algorithm to find the best path.
- Works within an Autonomous System (AS).
- Designed to overcome limitations of RIP (Routing Information Protocol).

Area in OSPF

- An OSPF area is a logical grouping of routers.
- Purpose: Reduce routing table size, minimize SPF calculations, provide scalability.
- Example: Area 0 → Backbone area (mandatory), Area 1, Area 2 → Non-backbone areas connected to Area 0.

Area Border Routers (ABR)

- A router that connects two or more areas.
- Belongs to backbone area (0) and at least one other area.
- Summarizes routes between areas.

Backbone Routers

- Any router that belongs to Area 0.
- Responsible for forwarding routing info inside the backbone.
- ABRs are always backbone routers (since they connect to Area 0).

Area Identification

- Each OSPF area has a unique Area ID (32-bit number).
- Can be written as decimal (e.g., 1) or IP address format (e.g., 0.0.0.1).

Disadvantages of RIP Protocol

- Hop-count limit: Maximum 15 hops → not suitable for large networks.
- Slow convergence: Takes time to detect failures.
- Count-to-infinity problem: Routing loops take long to resolve.
- Does not support VLSM (Variable Length Subnet Mask).

Remedies (Why OSPF is Better)

- OSPF supports unlimited hops.
- Faster convergence due to link-state updates.
- No count-to-infinity problem.
- Supports VLSM & CIDR.
- Uses cost (bandwidth-based) metric instead of hop count.

Features of OSPF

- Link-state protocol.
- Supports hierarchical design with areas.
- Uses cost as a metric.
- Fast convergence.
- Supports VLSM & CIDR.
- Authenticates routing updates.
- Scalability → works well in large networks.

Types of Links in OSPF

- Point-to-Point Link → between two routers.
- Broadcast Link → e.g., Ethernet LAN, supports DR/BDR election.
- Non-Broadcast Multi-Access (NBMA) Link → e.g., Frame Relay.
- Virtual Link → logical connection through Area 0.

Virtual Link

- A logical connection between two routers not directly connected to Area 0.
- Used when an area cannot connect directly to the backbone.
- Helps maintain hierarchical structure.

Link State Advertisements (LSA)

- OSPF routers exchange information using LSAs.
- Each LSA describes a part of the network topology.
- Stored in the Link State Database (LSDB).

OSPF Packet Types

- Hello Packet → Discover and maintain neighbors.
- Database Description (DBD) Packet → Summary of LSDB.
- Link State Request (LSR) Packet → Request for LSAs.
- Link State Update (LSU) Packet → Carry LSAs.
- Link State Acknowledgement (LSAck) Packet → Confirm receipt of LSAs.

Link State Update Packet

- Used to send LSAs between routers.
- Ensures routers have synchronized LSDBs.

General LSA Types

- Type 1 → Router LSA
- Type 2 → Network LSA
- Type 3 → Summary LSA (Network)
- Type 4 → Summary LSA (ASBR)
- Type 5 → External LSA

Router Link LSA (Type 1)

- Generated by every router.
- Describes its interfaces & neighbors.
- Flooded within the same area only.

Router Link Packet

- Another name for Router LSA (Type 1).
- Defines connections from the router to its directly connected networks.

Network Link LSA (Type 2)

- Generated by the Designated Router (DR).
- Represents all routers connected to a broadcast/multi-access network.
- Helps reduce flooding in large LANs.

Summary Link to Network LSA (Type 3)

- Generated by ABRs.
- Summarizes network routes from one area to another.

Summary Link to AS Boundary Router LSA (Type 4)

- Generated by ABRs.
- Advertises routes to AS Boundary Routers (ASBRs).